

1. What is an Operating System?

An operating system is software that helps you use the computer.
It connects the user, programs, and hardware.

Main functions:

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|---------------------------|----------------------------|
| 1. Runs programs | 2. Manages memory. |
| 3. Handles files | 4. Controls devices. |
| 5. Keeps the system safe. | 6. Gives a user interface. |

2. Differentiate between system software and application software:

System software runs the computer and helps other programs work (like Windows).
Application software helps you do tasks (like Word or games).

3. Main components of an Operating System:

Kernel: The main part that controls everything.

Memory manager: Handles computer memory.

File system: Manages files and folders.

Device manager: Controls hardware devices.

User interface: Let users talk to the computer.

4. Linux vs Windows:

Linux is free and open.

Windows is paid and closed.

Linux is safer and used by programmers.

Windows is easier for normal users.

5. Multithreading and Multiprocessing:

Multithreading means doing many small tasks at once inside one program.

Multiprocessing means using many processors to run different programs at once.

6. Role of OS in Cloud Computing:

The OS controls and shares cloud resources.

It manages memory, security, and connections between users and cloud servers.

It helps many people use the same system safely and smoothly.

